Refer to uploaded document:

- **Notes - Scaling and Transformations.pdf**

So far : Scaling transformations.

log

10 → 10^1 $\xrightarrow{\log}$ 1

100 → 10^2 → 2

1000 → 10^3 → 3

X → Skew (right) → log → drastically reduces skew.

→ Handles the outliers gracefully.

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots$

Gradient Descent Method (Search).

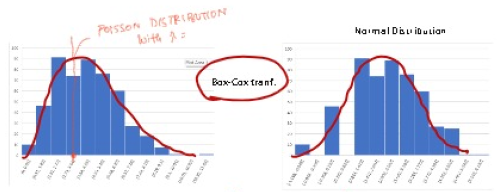
Box-Cox Transform

Original Y → transformed Y'

Box-Cox Transform $X(\lambda) = \begin{cases} \frac{x^{\lambda}-1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(x) & \text{if } \lambda = 0 \end{cases}$ ① Removes 'Skew' from the data ② Makes the variance 'uniform'

Iterative process

"Maximum Likelihood Estimation"



Y $\xrightarrow{\text{Box-Cox}}$ Y'

Original space ← Predicted Value.

- Box-Cox transformation is primarily applied to the dependent variable to stabilize variance and normalize residuals.
- It helps improve linear regression models when normality and homoscedasticity assumptions are violated.
- **It is not commonly applied to independent variables**, since transformations of predictors are typically handled differently (e.g., log)

When the Box-Cox transformation is applied to the dependent variable (response variable), the model's predictions will be in the transformed space. To interpret them correctly, **we need to inverse transform the predictions back to the original scale.**

If the Box-Cox transformation was applied using:

$$y' = \begin{cases} \frac{y^{\lambda}-1}{\lambda}, & \text{if } \lambda \neq 0 \\ \ln(y), & \text{if } \lambda = 0 \end{cases}$$

value of λ is iteratively established. M.L.E.

Then, the inverse transformation (to convert back to the original space) is:

$$y = \begin{cases} (\lambda y' + 1)^{\frac{1}{\lambda}}, & \text{if } \lambda \neq 0 \\ e^{y'}, & \text{if } \lambda = 0 \end{cases}$$

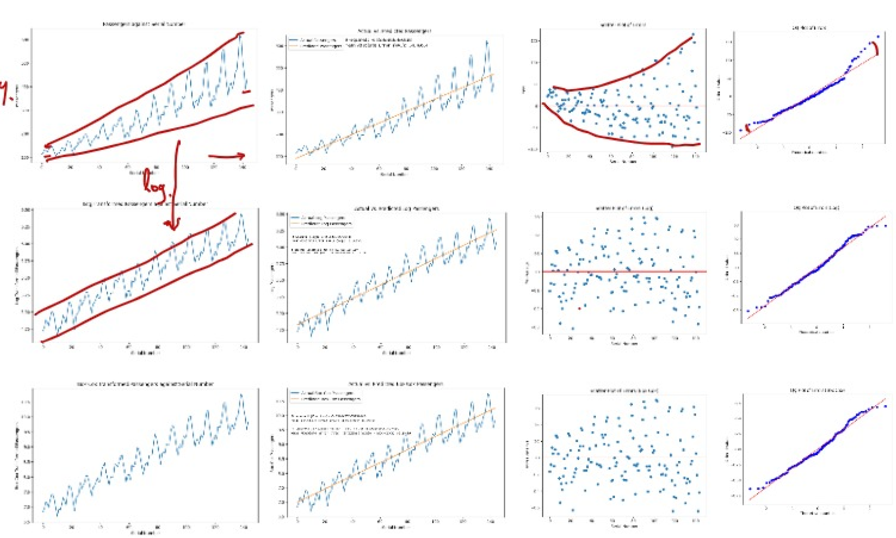
Application of transforms and the impact: Example - Airline Passengers' Data

The following plots illustrate the effect of log transformation and Box-Cox transformations (of Y) on Regression errors and results. Particularly observe the following:

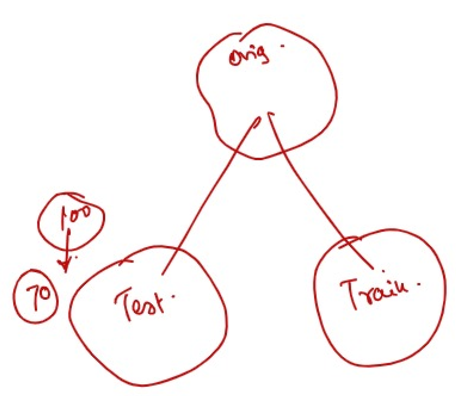
- Changes in the shape of the line-plot of the airline passengers data
- Changes in the error scatter plots: in the second and third cases the errors appear more random; in the first case the errors seem to have a predictable trend, in line with the increasing variance in the data set with time
- The QQ-plots, indicate better 'normal distribution' of the errors in the second and the third cases. The red-line in the QQ-plot represents perfect normal distribution - any deviation of actual data (in this case, errors) from the red line indicates departure from the normal distribution.
- Observe the R2 and MAE values. Regression model created using Box-Cox transformed 'Y' data shows improvement (even in the original space).

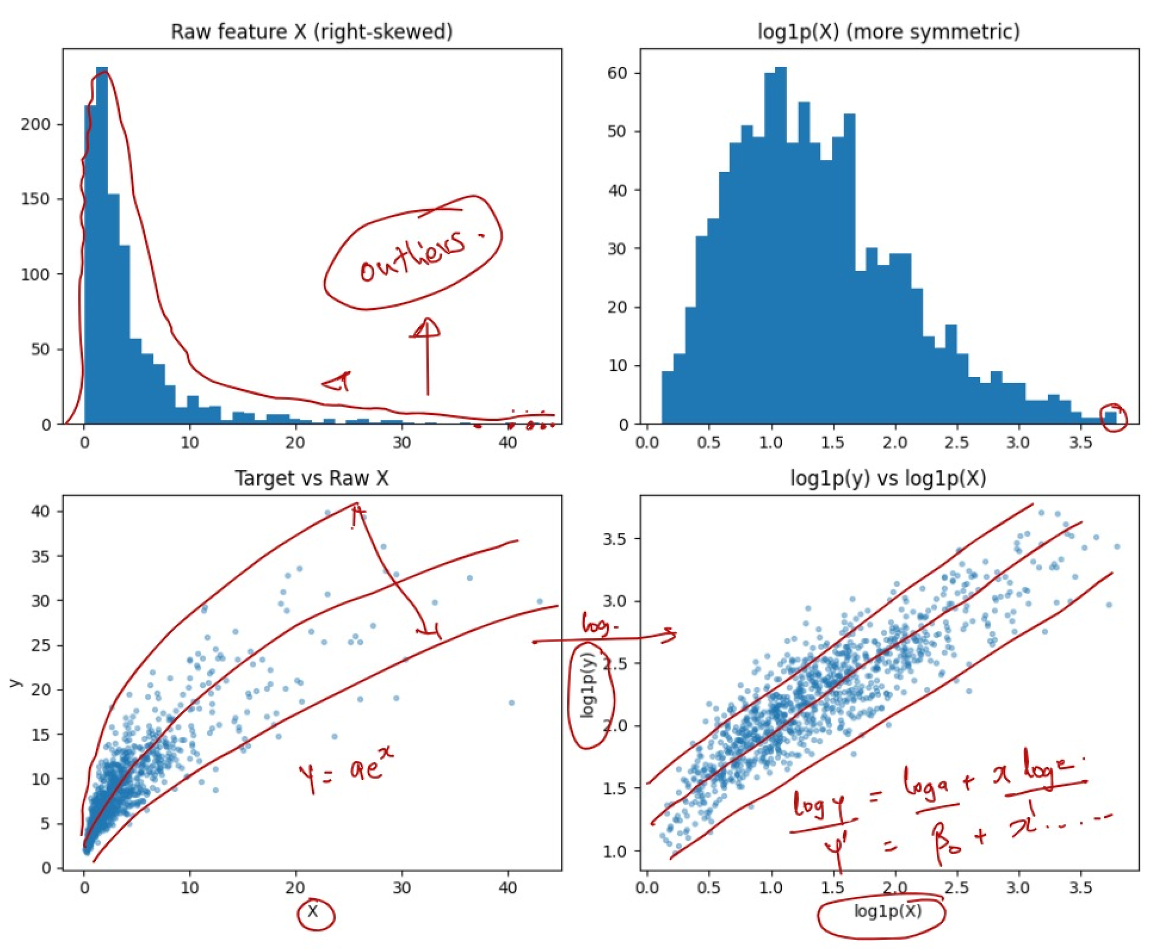
changing Variance → heteroscedasticity.

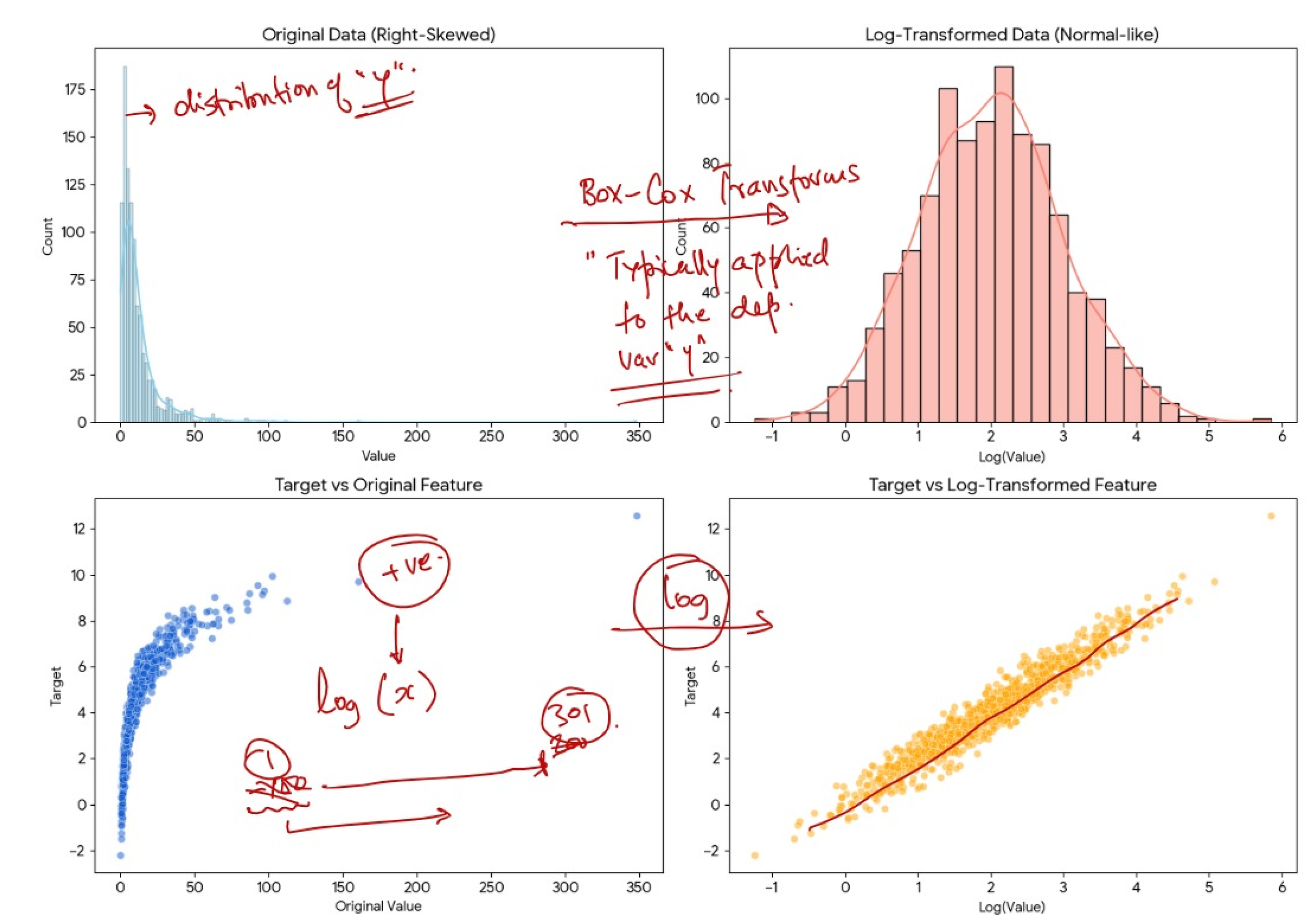
Many algos assume → Variance is constant.



Pop.







Box-Cox transformation and log transforms can be applied to **features** X and/or **target** y (depending on modeling goal). They are commonly used to help with:

1. **Reducing skewness** (right-skew mainly; left-skew with other methods or reflection)
2. **Reducing influence of extreme large values** (not "removing outliers")
3. **Mitigating heteroscedasticity** (non-constant variance)
4. **Making nonlinear relationships more linear**
5. **Improving model assumptions** for linear/statistical models
6. **Stabilizing optimization / training** for some ML algorithms
7. **Improving interpretability** in some log models (% effects, elasticity)

$x_{21} \ldots x_{2,100}$

Normally. $Y = f(X)$ ← apply various transforms (Scaling, log, Box-Cox)

no transf. used for training

new observation $(x_1, x_2, x_3 \ldots)$

original space. $\hat{Y} \leftarrow f$

When Y is transformed

original Y → Box-Cox (λ=3.6) → Y'

X → various transforms → X'

$Y' \leftarrow f(x_1' x_2' x_3')$ predicted.

Y (original) ← inverse of Box-Cox (λ=3.6) ← Y'

Original Space. transformed space.

Parametric Methods:

Linear Regr. $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots$

Non-Parametric Methods: KNN-Regression.

Able to predict

Algorithms work well if the predicted var is normally distributed.

height →

weight →

Any quantity that depends on a number of factors ⇒ shows normal distribution.